Practical 1 : Salesforce Apex Programming Practical Guide

Complete Beginner Friendly Step-by-Step Practical Guide for Apex Programming

This practical explains:

1. Creating Salesforce Developer Account
2. Executing Linear Search Program using Apex
3. Executing Binary Search Program using Apex

Complete Beginner Friendly Step-by-Step Practical Guide

This guide explains:

1. Creating Salesforce Developer Account
2. Executing Linear Search Program using Apex
3. Executing Binary Search Program using Apex
4. Creating a Mini Project in Salesforce Cloud

PART 1 : Introduction to Salesforce

Salesforce is a cloud computing platform used for:

- CRM (Customer Relationship Management)
- Cloud Application Development
- Database Management
- Automation
- Apex Programming
- Enterprise Applications

Apex is the programming language used in Salesforce.

It is similar to Java programming language.

PART 2 : Create Salesforce Developer Account

Step 1 : Open Salesforce Developer Signup Website

Open the official Salesforce Developer signup page:

https://developer.salesforce.com/signup

Step 2 : Fill Registration Details

Fill all required information carefully.

Field	Example
First Name	Mayank
Last Name	Vispute
Email	Your Email
Company	Your College Name
Country	India
Username	Your Email Format Username

Example Username:

mayankvispute@agentforce.com

Note:
Username must look like an email address.

Step 3 : Verify Email

After signup:

1. Open your email inbox
2. Open Salesforce verification mail
3. Click:

RESET PASSWORD

4. Create your password
5. Save username and password carefully

PART 3 : Login to Salesforce

Open:

https://login.salesforce.com

Enter:

- Username
- Password

Click:

Login

PART 4 : Open Developer Console

Developer Console is used to write and execute Apex programs.

Step 1

After login:

Go to the top-right corner.

Click:

Gear Icon (⚙)

Step 2

Click:

Developer Console

A new window will open.

This is called the Developer Console.

PART 5 : Linear Search Program in Apex

Theory

Linear Search checks elements one by one until the element is found.

Example:

10 → 20 → 30 → 40 → 50

If searching for 50:

The program checks:

10 → 20 → 30 → 40 → 50

Then stops.

Step-by-Step Execution

Step 1 : Create Apex Class

Inside Developer Console:

File
↓
New
↓
Apex Class

Step 2 : Enter Class Name

Enter:

LinearSearchExample

Click:

OK

Step 3 : Delete Default Code

Delete all generated sample code.

Step 4 : Paste Linear Search Program

```
public class LinearSearchExample {

    // Linear Search Method
    public static Integer linearSearch(List<Integer> arr, Integer target) {

        // Loop through list
        for(Integer i = 0; i < arr.size(); i++) {

            // Check element
            if(arr[i] == target) {

                return i;
            }
        }

        // Element not found
        return -1;
    }

    // Main Method
    public static void mainMethod() {

        // Creating list
        List<Integer> numbers = new List<Integer>{10,20,30,40,50};

        // Target value
        Integer target = 50;

        // Calling method
        Integer result = linearSearch(numbers, target);

        // Display output
        if(result != -1) {

            System.debug('Element found at index: ' + result);

        } else {

            System.debug('Element not found');
        }
    }
}
```

Step 5 : Save Program

Press:

Ctrl + S

OR

File → Save

Step 6 : Open Execute Anonymous Window

Inside Developer Console:

Debug
↓
Open Execute Anonymous Window

Step 7 : Execute Program

Inside popup window enter:

```
LinearSearchExample.mainMethod();
```

Step 8 : Open Log

Tick:

Open Log

Click:

Execute

Step 9 : View Output

Inside Logs:

Tick:

Debug Only

Final Output:

Element found at index: 4

Viva Questions for Linear Search

Q1 What is Linear Search?

Linear Search checks elements one by one.

Q2 Time Complexity of Linear Search?

O(n)

Q3 Advantage of Linear Search?

Simple and easy to implement.

Q4 Disadvantage of Linear Search?

Slow for large datasets.

PART 6 : Binary Search Program in Apex

Theory

Binary Search works on sorted data.

It divides the array into halves repeatedly.

It is faster than Linear Search.

Example

Sorted Array:

5 → 10 → 15 → 20 → 25 → 30 → 35

Searching for:

30

Binary Search checks middle element first.

Step-by-Step Execution

Step 1 : Create Apex Class

Inside Developer Console:

File
↓
New
↓
Apex Class

Step 2 : Enter Class Name

Enter:

BinarySearchExample

Click:

OK

Step 3 : Delete Default Code

Delete all generated sample code.

Step 4 : Paste Binary Search Program

```
public class BinarySearchExample {

    // Binary Search Method
    public static Integer binarySearch(List<Integer> arr, Integer target) {
```

```
        Integer low = 0;
        Integer high = arr.size() - 1;

        while(low <= high) {

            Integer mid = (low + high) / 2;

            // Element found
            if(arr[mid] == target) {

                return mid;
            }

            // Search right side
            else if(arr[mid] < target) {

                low = mid + 1;
            }

            // Search left side
            else {

                high = mid - 1;
            }
        }

        // Element not found
        return -1;
    }

    // Main Method
    public static void mainMethod() {

        List<Integer> numbers =
            new List<Integer>{5,10,15,20,25,30,35,40};

        Integer target = 30;

        Integer result = binarySearch(numbers, target);

        if(result != -1) {

            System.debug('Element found at index: ' + result);

        } else {

            System.debug('Element not found');
        }
    }
}
```

Step 5 : Save Program

Press:

Ctrl + S

Step 6 : Open Execute Anonymous Window

Debug
↓
Open Execute Anonymous Window

Step 7 : Execute Program

Enter:

```
BinarySearchExample.mainMethod();
```

Tick:

Open Log

Click:

Execute

Step 8 : View Output

Tick:

Debug Only

Final Output:

Element found at index: 5

Viva Questions for Binary Search

Q1 What is Binary Search?

Binary Search divides the array into halves repeatedly.

Q2 Binary Search works on?

Sorted data only.

Q3 Time Complexity of Binary Search?

O(log n)

Q4 Why Binary Search is faster?

Because it divides the data into halves.

PART 7 : Difference Between Linear Search and Binary Search

Linear Search	Binary Search
Sequential Search	Divide and Conquer
Slower	Faster
Works on any data	Works only on sorted data
Time Complexity O(n)	Time Complexity O(log n)

==========

==========

==========

==========

PRACTICAL 2 : Salesforce Mini Project Development Guide

==========

==========

Complete Beginner Friendly Step-by-Step Guide for Salesforce Mini Project

PART 8 : Mini Project in Salesforce Cloud

Project Title

Student Management System

Objective

To create a custom Salesforce application for managing student records.

Features

The project stores:

- Student Name
- Roll Number
- Course
- Mobile Number

Step-by-Step Mini Project Development

Step 1 : Open Setup

After login:

Go to top-right corner.

Click:

Gear Icon (⚙)
↓
Setup

Step 2 : Open Object Manager

At top menu click:

Object Manager

Step 3 : Create Custom Object

Click:

Create
↓
Custom Object

Step 4 : Enter Object Details

Field	Value
Label	Student
Plural Label	Students
Object Name	Student

Tick:

Launch New Custom Tab Wizard

Click:

Save

Step 5 : Add Tab Visibility

Keep:

Default On

Click:

Next

Then click:

Save

PART 9 : Create Fields

Step 1 : Open Fields & Relationships

Inside Student Object:

Click:

Fields & Relationships

Step 2 : Create Student Name Field

Click:

New

Choose:

Text

Click:

Next

Enter:

Field Label = Student Name

Click:

Next
↓
Next
↓
Save

Step 3 : Create Roll Number Field

Click:

New

Choose:

Number

Enter:

Field	Value
Field Label	Roll Number
Length	5

Click:

Next
↓
Next
↓
Save

Step 4 : Create Course Field

Click:

New

Choose:

Text

Enter:

Field Label = Course

Click:

Next
↓
Next
↓
Save

Step 5 : Create Mobile Number Field

Click:

New

Choose:

Phone

Enter:

Field Label = Mobile Number

Click:

Next
↓
Next
↓
Save

PART 10 : Add Student Records

Step 1 : Open App Launcher

Click:

9 Dots Icon

Step 2 : Search Students

Search:

Students

Open it.

Step 3 : Add New Record

Click:

New

Step 4 : Enter Student Details

Student Name	Roll Number	Course	Mobile Number
Rahul	101	Computer	9876543210
Priya	102	IT	9988776655

Click:

Save

Viva Questions for Mini Project

Q1 What is Salesforce?

Salesforce is a cloud-based CRM platform.

Q2 What is Apex?

Apex is the programming language used in Salesforce.

Q3 What is a Custom Object?

A custom object is a user-created database table in Salesforce.

Q4 What is a Field?

A field is a column used to store data.

Q5 What is CRM?

CRM stands for Customer Relationship Management.

Q6 Difference Between Standard Object and Custom Object?

Standard Object	Custom Object
Predefined by Salesforce	Created by User

PART 11 : Final Practical Flow

For Apex Programs

Login
↓
Developer Console
↓
New Apex Class
↓
Paste Program
↓
Save
↓
Execute Anonymous Window
↓
Run Program
↓
View Logs

For Mini Project

Setup
↓
Object Manager
↓
Create Custom Object
↓
Create Fields
↓
Create Tab
↓
Add Records

Final Conclusion

In this Salesforce Cloud Computing practical we successfully learned:

- Creating Salesforce Developer Account
- Opening Developer Console
- Writing Apex Programs
- Executing Linear Search Program
- Executing Binary Search Program
- Creating a Salesforce Mini Project
- Creating Custom Objects and Fields
- Managing Records in Salesforce

Salesforce provides a powerful cloud platform for application development, database management, automation, and enterprise solutions using Apex programming language.

Final Exam Tips

- Carry Salesforce Username and Password
- Save Apex Classes properly
- Always use Execute Anonymous Window for execution
- Tick Open Log before execution
- Use Debug Only for clean output
- Binary Search requires sorted data
- Practice once before examination
- Speak confidently during viva